# Xuenan Li

Department of Mathematics
Purdue University
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## Research Interests

Calculus of Variations, Partial Differential Equations and Spectral Theory
with particular focus on problems from physics, mechanics, and materials science

New Methods for Designing Rigid Frameworks with higher-order rigidity
including an optimization-based approach to finding new tensegrity structures

## Academic Position

2026- Assistant Professor
Department of Mathematics, Purdue University

2023-2026 Chu Assistant Professor in Applied Mathematics
Columbia University
*Mentor: Michael I. Weinstein*

## Education

2018- 2023 Ph.D. in Mathematics
Courant Institute of Mathematical Sciences, New York University, NY, USA
*Advisor: Robert V. Kohn*
*Thesis: Guest-Hutchinson modes and the macroscopic energy of the Kagome lattice metamaterial and related systems*

2015-2018 B.S. in Mathematics and Data Science (double major)
University of Michigan, Ann Arbor, MI, USA
*Advisors: Charles R. Doering and Robert Krasny*

## Publications & Preprints

1. **Li, Xuenan** and Kohn, Robert V. (2023) *Some results on the Guest-Hutchinson modes and periodic mechanisms of the Kagome lattice metamaterial,* J Mech Phys Solids 178 (Sept 2023) 105311, published version, arXiv:2210.00382 (55 pages).

2. **Li, Xuenan**, Mihnea Leonte, Santangelo, Christian D., and Holmes-Cerfon, Miranda (2025) *A Constrained Saddle Search Approach for Constructing Singular and Flexible Bar Frameworks,* accepted by 3rd IMA Conference on Mathematics of Robotics (peer reviewed), published version, arXiv:2503.14807 (9 pages).

3. **Li, Xuenan** and Kohn, Robert V. (2026) *The effective energy of a lattice metamaterial,* Journal of Elasticity 158 (2026), published version, arXiv:2505.05436.

4. **Li, Xuenan** and Kohn, Robert V. (2025) *A nonlinear homogenization-based perspective on the soft modes and effective energies of some conformal metamaterials,* arXiv:2509.16907 (54 pages).

5. **Li, Xuenan**, Santangelo, Christian D., and Holmes-Cerfon, Miranda (2025) *A Constrained Optimization Approach for Constructing Rigid Bar Frameworks with Higher-order Rigidity,* submitted, arXiv:2509.23072 (31 pages).
6. **Li, Xuenan**, Liu, Chun, and Qi, Di (2025) *An Asymptotic Approach for Modeling Multiscale Complex Fluids at the Fast Relaxation Limit,* submitted, arXiv:2512.17134 (37 pages).
7. **Li, Xuenan** and Weinstein, Michael I. (2026) *Pseudo-magnetism in the slowly-strained discrete honeycomb lattice,* arXiv:2603.27503 (91 pages).

## Teaching Experience

Spring 2026 — *Instructor*
MA 34100 Foundations Of Analysis (Section 65 & 67)
Purdue University

Spring 2026, Spring 2025, Spring 2024 — *Instructor*
APAM E4101 Dynamical systems
Columbia University

Fall 2025, Fall 2024, Fall 2023 — *Instructor*
APAM E4204 Functions of a complex variable
Columbia University

Fall 2022 — *Teaching Assistant*
MATH-UA.0262-007 Ordinary Differential Equations
New York University

Spring 2022 — *Teaching Assistant*
MATH-UA.0140-007: Linear Algebra
New York University

Fall 2021 — *Teaching Assistant*
MATH-UA.0263-002: Partial Differential Equations
New York University

Spring 2021 — *Teaching Assistant*
MATH-UA.0121-017: Calculus I
New York University

## Conferences & Workshop Talks

10/2026 — *Pseudo-magnetism in a strained discrete honeycomb lattice*, SIAM New York-New Jersey-Pennsylvania Section Annual Meeting , New Brunswick, NJ, October 24 - October 25, 2026.

10/2026 — *Pseudo-magnetism in a strained discrete honeycomb lattice (poster)*, Simons Collaboration on Extreme Wave Phenomena Based on Symmetries Annual Meeting, New York, NY, October 22- October 23, 2026.

10/2026 — *Pseudo-magnetism in a strained discrete honeycomb lattice*, NSF RTG Conference on Materials and Optimization, Salt Lake City, UT, October 16 - October 18, 2026.

10/2026 *Pseudo-magnetism in a strained discrete honeycomb lattice (poster)*, Frontiers of Mathematical and Computational Aspects of Materials Science, Carnegie Mellon University, Pittsburgh, PA, October 12 -14, 2026.

9/2026 *Pseudo-magnetism in a strained discrete honeycomb lattice*, SIAM Great Lakes Section Annual Meeting, Purdue University, IN, September 19 - September 20, 2026.

6/2026 *Pseudo-magnetism in a strained discrete honeycomb lattice (poster)*, Mathematical Aspects of 2D Quantum Materials and Meta-materials, Institute for Mathematical and Statistical Innovation (IMSI), Chicago, IL, June 8 - June 11, 2026.

6/2026 *Soft modes in mechanism-based mechanical metamaterials*, Geometric PDE Challenges in Science and Engineering, Brin Mathematics Research Center, College Park, MD, June 2-June 5, 2026.

5/2026 *Pseudo-magnetism in a strained discrete honeycomb lattice (poster)*, Moire Materials Workshop at Simons Foundation, New York, NY, May 11-13, 2026.

11/2025 *Pseudo-magnetism in a strained discrete honeycomb lattice*, SIAM Conference on Analysis of Partial Differential Equations (PD25), Pittsburgh, PA, Nov 17, 2025.

10/2025 *Pseudo-magnetism in a strained discrete honeycomb lattice*, 2025 Fall Central Sectional Meeting, St. Louis, MO, Oct 18, 2025.

09/2025 *A Constrained Saddle Search Approach for Constructing Singular and Flexible Frameworks*, 3rd IMA Conference on Mathematics of Robotics, Manchester, UK, Sep 25, 2025.

06/2025 *Edge states in the discrete slowly-varying Su–Schrieffer–Heeger (SSH) model*, SIAM Annual Meeting (AN25), Montreal, July 28, 2025.

06/2025 *Edge states in the discrete slowly-varying Su–Schrieffer–Heeger (SSH) model*, International Conference on Elastic, Electrical, Transport, and Optical Properties of Inhomogeneous Media (ETOPIM 13), New York, June 16, 2025.

05/2025 *A Constrained Optimization Approach for Constructing Rigid Bar Frameworks with Higher-Order Rigidity* (poster), NSF CompMath Meeting 2025, University of Utah, May 8-9, 2025.

05/2025 *A Constrained Optimization Approach for Constructing Bar Frameworks with Higher-order Rigidity*, Sayas Numerics Day 2025, University of Delaware, May 3, 2025.

04/2025 *A Constrained Optimization Approach for Constructing Tensegrity Frameworks*, East Coast Optimization Meeting 2025, April 18, 2025.

04/2025 *Deformations in soft mechanical metamaterials:modeling, analysis, and applications*, Workshop for Women in PDE and Applied Mathematics in Northeast Region, University of Connecticut, April 12, 2025.

04/2025 *A Constrained Optimization Approach for Constructing Singular Bar Frameworks*, ICERM Geometry of Materials Workshop, April 08, 2025.

04/2025 *Edge states in the discrete slowly-varying Su–Schrieffer–Heeger model*, AMS Spring Eastern Sectional Meeting, April 6, 2025.

06/2024 *The Kagome Lattice as a Mechanism-based Mechanical Metamaterial* (poster), Mathematics of Multiscale and Multiphysics Phenomena in Materials Science, Banff, Calgory, June 16 - 21, 2024.

05/2024 *The Kagome Lattice as a Mechanism-basedMechanical Metamaterial*, SIAM Conference on Mathematical Aspects of Materials Science (MS24), Pittsburgh, Pennsylvania, May 19–23 , 2024.

10/2023 *The Kagome lattice as a mechanical metamaterial*, 2023 SES Annual Technical Meeting (SES 2023), Saint Paul, Minnesota, October 8-11, 2023.

8/2023 *The macroscopic behavior of the Kagome lattice metamaterial*, 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023), Tokyo, August 20-25, 2023.

10/2022 *Some results on the Guest-Hutchinson modes and periodic mechanisms of the Kagome lattice metamaterial* (poster), Simons Collaboration on Extreme Wave Phenomena Based on Symmetries Annual Meeting, Flatiron Institute, New York, October 20–21, 2022.

10/2022 *The macroscopic energy of the Kagome lattice metamaterials*, UpState NY Soft Matter Symposium, Rochester Institute of Technology, October 11, 2022.

7/2022 *Kagome lattice as a mechanical metamaterial*, SIAM Annual Meeting (AN22), Pittsburgh, July 11-15, 2022.

6/2022 *The Kagome lattice as a mechanical metamaterial* (poster), UMass Summer School on Soft Solids and Complex Fluids, University of Massachusetts Amherst, June 5–9, 2022.

10/2021 *A mathematical perspective on Guest-Hutchinson modes* (poster), Simons Collaboration on Extreme Wave Phenomena Based on Symmetries Annual Meeting, Flatiron Institute, New York, October 21–22, 2021.

## Invited Colloquium & Seminar Talks

9/2026 *Pseudo-magnetism in a strained discrete honeycomb lattice*, Mathematical Physics Seminar, Purdue University, West Lafayette, IN, September 30, 2026.

9/2026 *Pseudo-magnetism in a strained discrete honeycomb lattice*, Applied & Interdisciplinary Mathematics, University of Michigan, Ann Arbor, MI, September 11, 2026.

9/2026 *Pseudo-magnetism in a strained discrete honeycomb lattice*, Applied and Computational Mathematics Seminar, Auburn University, Auburn, AL, September 4, 2026.

6/2026 *Soft modes in mechanism-based mechanical metamaterials*, Special Physics Seminar, University of Pennsylvania, June 22, 2026.

11/2025 *Soft modes in mechanism-based mechanical metamaterials:modeling, analysis, and applications*, PDE seminar, Purdue University, November 7, 2025.

11/2025 *Soft modes in mechanism-based mechanical metamaterials:modeling, analysis, and applications*, Numerical analysis and PDE seminar, University of Delaware, November 3, 2025.

10/2025 *Soft modes in mechanism-based mechanical metamaterials:modeling, analysis, and applications*, Applied analysis seminar, Louisiana State University, October 26, 2025.

03/2025 *Deformations in soft mechanical metamaterials:modeling, analysis, and applications*, Fluid Mechanics & Waves Seminar, New Jersey Institute of Technology, March 24, 2025.

11/2024 *Deformations in soft mechanical metamaterials:modeling, analysis, and applications*, Analysis & PDE Seminarat, Worcester Polytechnic Institute, November 14, 2024

10/2024 *Deformations in soft mechanical metamaterials:modeling, analysis, and applications*, PDE & Differential Geometry Seminar, University of Connecticut, October 28, 2024

2/2024 *The Kagome lattice as a mechanical metamaterial*, PDE Seminar, Arizona State University, Phoenix, February 2, 2024.

11/2023 *The Kagome lattice as a mechanical metamaterial*, Data Science Seminar, University of Minnesota Twin Cities, Saint Paul, Minnesota, November 27, 2023.

11/2023 *The Kagome lattice as a mechanical metamaterial*, Applied Math Seminar, The University of British Columbia, Vancouver, November 7, 2023.

1/2023 *The Kagome Lattice as a Mechanism-based Metamaterial*, Applied Math Colloquium, Columbia University, New York, January 9, 2023.

3/2022 *Some results on Guest-Hutchinson modes and periodic mechanisms of the Kagome lattice as a metamaterial*, Modeling Simulation Group Seminar, Courant Institute, New York, March 3, 2022.

10/2021 *The Kagome lattice as a mechanical metamaterial*, Analysis and Applied Mathematics Seminar, University of Illinois Chicago, Chicago, October 11, 2021.

## Academic Services

Spring 2026 Thesis defense committee member:
PhD Candidate: *Yan Cheng*, Department of Applied Physics and Applied Mathematics
Dissertation Title: *Computational Inverse Problems in Linear and Nonlinear Wave Propagation.*

Spring 2026 Thesis defense committee member:
PhD Candidate: *Yinxi Pan*, Department of Applied Physics and Applied Mathematics
Dissertation Title: *Inverse Problems and Model Reduction for Transport and Diffusion Models.*

Spring 2026 Thesis defense committee member:
PhD Candidate: *Daozhe Lin*, Department of Applied Physics and Applied Mathematics
Dissertation Title: *Nonlinear Operator Equations in Hilbert Spaces: Foundations, Optimal Recovery, and Active Learning.*

Fall 2025 *Co-organizer*, Advances in the Spectral Analysis of Schrödinger Operators in Complex Materials
SIAM Conference on Analysis of Partial Differential Equations (PD25)
Pittsburgh, Pennsylvania, U.S., November 17-20, 2025

Spring 2025 Thesis defense committee member:
PhD Candidate: *Edith Jin Zhang*, Department of Applied Physics and Applied Mathematics
Dissertation Title: *Ginzburg–Landau Functionals and Reaction–diffusion Equations in the Large-Graph Limit.*

Fall 2025 -present
Master Student Research Program at Columbia University with *Jingkai Wang*.
**Update**: Jingkai will be a PhD student in Mathematics at University of Santa Barbara starting Fall 2026.

Fall 2024 -Spring 2025
Master Student Research Program at Columbia University with *Wendi Lu*.
**Update**: Wendi is now a PhD student in Mathematics at University of Wisconsin-Madison started Fall 2025.

Fall 2024
Thesis defense committee member:
PhD Candidate: *Zirui Xu*, Department of Applied Physics and Applied Mathematics
Dissertation Title: *Ohta–Kawasaki Energy and Its Phase-Field Simulation.*

Fall 2024
*Co-organizer*, APAM Research Conference, Columbia University, New York, NY

Summer 2024
Undergraduate Student Research Program at Columbia University
with *Mihnea Leonte and Andrew Yang*;
Program title: *Designing singular structures in bar-joint networks*;
Funded by *Bonomi Scholars* at Columbia University.

**Update**: Mihnea Leonte received the Applied Mathematics Faculty Award of class 2025 and is a current Master student in Financial Engineering at Columbia University;

Andrew Yang received the Applied Physics Faculty Award of class 2025 and is now a PhD student in Applied Physics at California Institute of Technology.

Fall 2023 -Spring 2026
*Co-organizer*, Wave Working Group meetings, Columbia University, New York, NY

## Honors & Awards

2025
NSF-AWM travel grant

2023
SIAM Student Travel Award, 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023)

2022
Sandra Bleistein Prize, New York University

2022
SIAM Student Travel Award, SIAM Annual Meeting (AN22)

2018–2023
New York University MacCracken Graduate Scholarship, New York University

2018
Wilfred Kaplan Award in Applied Mathematics, University of Michigan

2018
Outstanding Graduating Senior Award, University of Michigan

2017–2018
James B. Angell Scholar, University of Michigan

2017
Sumner B.Myers Award in Analysis, University of Michigan
*This is awarded to the undergraduate student who is most excellent in the study of analysis.*